

Power

EEE 203

Passive Sign Convention

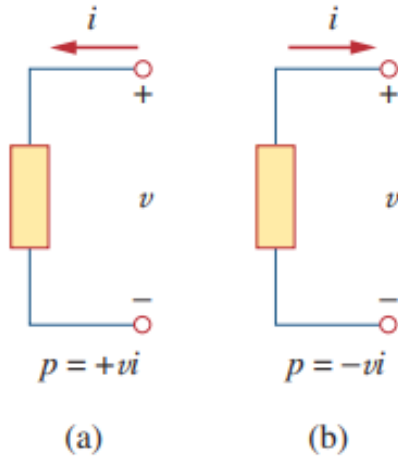


Figure 1.8

Reference polarities for power using the passive sign convention: (a) absorbing power, (b) supplying power.

Resistor	Absorbs Energy
Capacitor	Stores and Releases Energy
Inductor	Stores and Releases Energy

First Order RC Circuit

- Capacitor stores energy in the electric field
- Rise of the electric field during storing phase

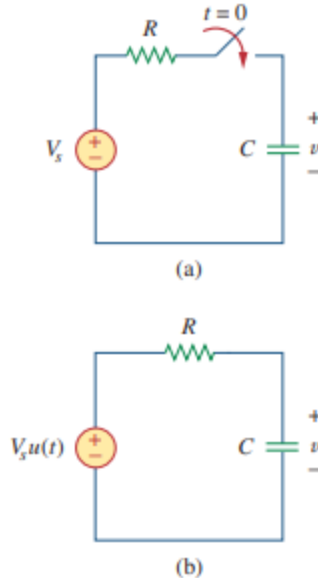
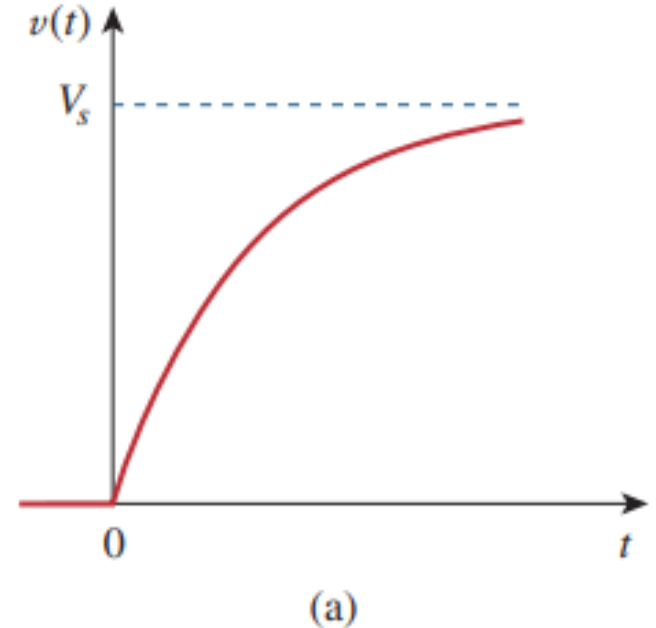


Figure 7.40
An RC circuit with voltage step input.



First Order RC Circuit

- Capacitor stores energy in the electric field
- Collapse of the electric field during releasing phase

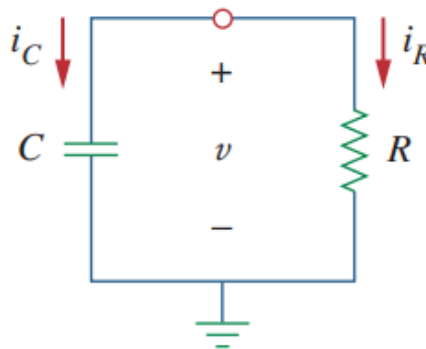


Figure 7.1
A source-free RC circuit.

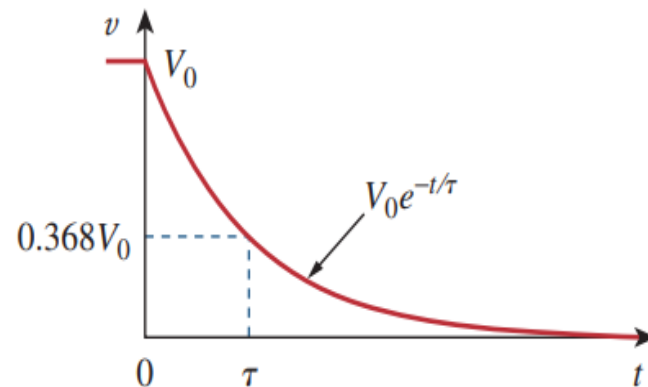


Figure 7.2
The voltage response of the RC circuit.

Instantaneous Power

- The power delivered to a load **at any instant** is defined by the product of the applied voltage and the resulting current:

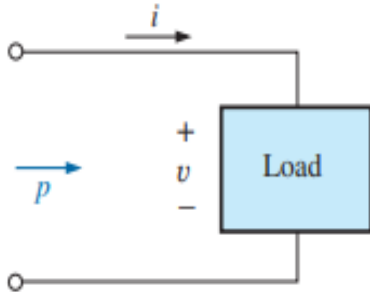


FIG. 1

Defining the power delivered to a load.

Voltage

$$v = V_m \sin(\omega t + \theta)$$

Current

$$i = I_m \sin \omega t$$

Instantaneous
Power

$$p = v i$$

Load Type	θ (degree)
Resistive Load	0
Pure Inductive Load	90
Pure Capacitive Load	-90
Primarily Inductive	+ve (somewhere between 0 and 90)
Primarily Capacitive	-ve (somewhere between 0 and -90)

Instantaneous Power

Substituting the above equations for v and i into the power equation results in

$$p = V_m I_m \sin \omega t \sin(\omega t + \theta)$$

If we now apply a number of trigonometric identities, the following form for the power equation results:

$$p = VI \cos \theta (1 - \cos 2\omega t) + VI \sin \theta \sin(2\omega t)$$

where V and I are the rms values.

Resistive Circuit

- For a purely resistive circuit (such as that in Fig. 2), v and i are in phase, and θ is 0° .
- The total power delivered to a resistor will be dissipated in the form of heat.

$$p = VI \cos \theta (1 - \cos 2\omega t) + VI \sin \theta \sin(2\omega t)$$

Remains

Vanishes

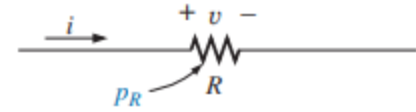


FIG. 2

Determining the power delivered to a purely resistive load.

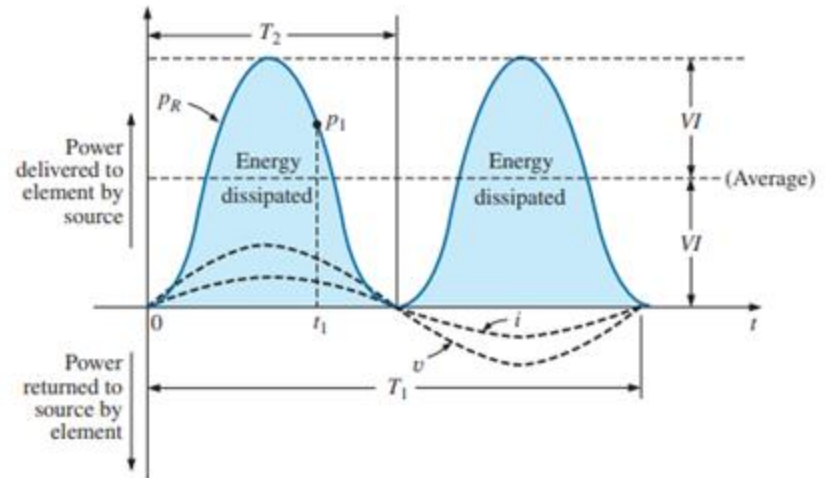


FIG. 3

Power versus time for a purely resistive load.

Note: V and I are the rms values

Inductive Circuit

- For a purely inductive circuit (such as that in Fig. 8), v leads i by 90° , as shown in Fig. 9. θ is 90° .
- The net flow of power to the pure (ideal) inductor is zero over a full cycle, and no energy is lost in the transaction.

$$p = VI \cos \theta (1 - \cos 2\omega t) + VI \sin \theta \sin(2\omega t)$$

Vanishes

Remains

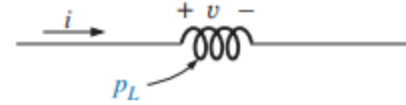


FIG. 8

Defining the power level for a purely inductive load.

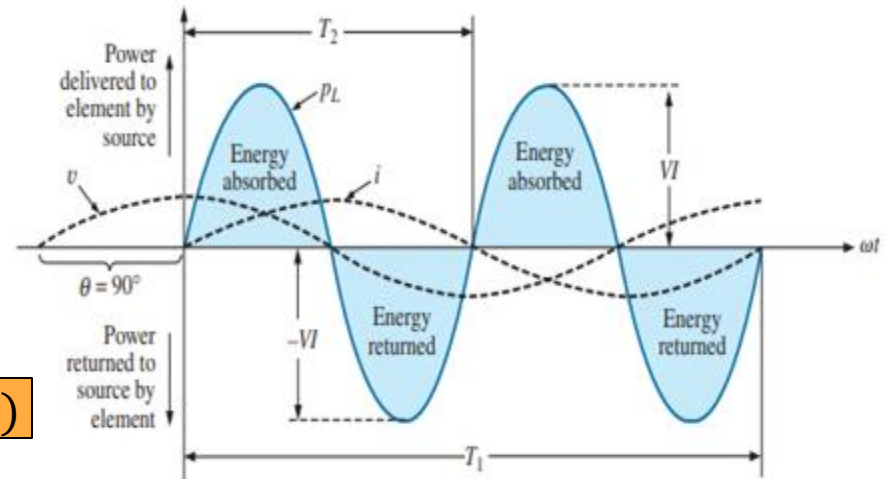


FIG. 9

The power curve for a purely inductive load.

Capacitive Circuit

- For a purely capacitive circuit (such as that in Fig. 12), i leads v by 90° , as shown in Fig. 13. θ is -90° .
- The net flow of power to the pure (ideal) capacitor is zero over a full cycle.

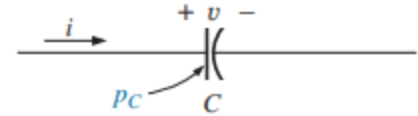


FIG. 12

Defining the power level for a purely capacitive load.

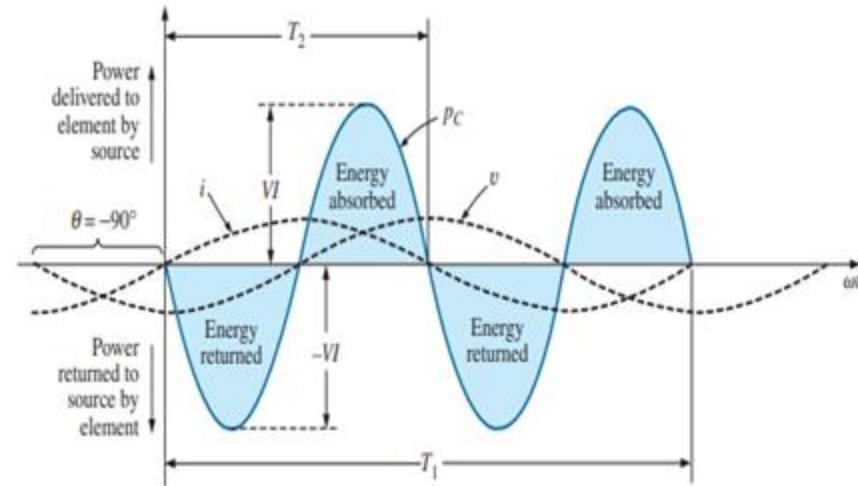


FIG. 13

The power curve for a purely capacitive load.

$$p = VI \cos \theta (1 - \cos 2\omega t) + VI \sin \theta \sin(2\omega t)$$

Vanishes

Remains

Note: V and I are the rms values

General Load

Load comprises resistive, inductive, and capacitive components.

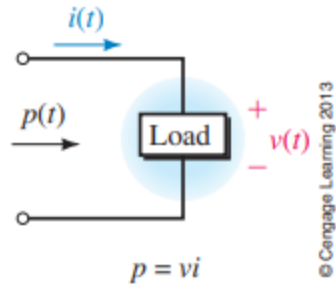


FIGURE 17-1 Voltage, current, and power references. When p is positive, power is in the direction of the reference arrow.

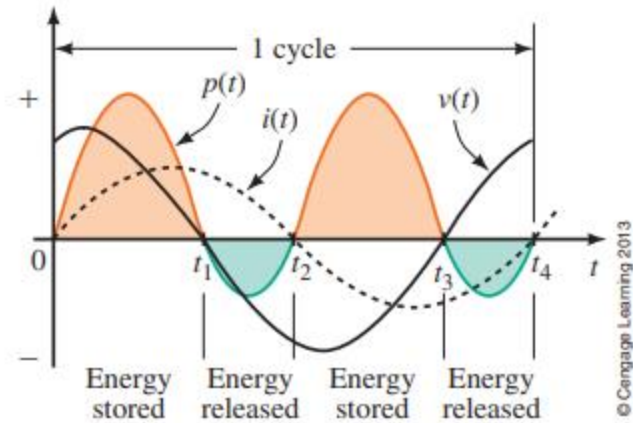


FIGURE 17-2 Instantaneous power in an ac circuit. Positive p represents power to the load; negative p represents power returned from the load.

Active Power

- Active Power is the average power flowing to the load.
- Active Power is denoted by the letter P , and its unit of measure is the watts (W). If P is positive, then, on average, more power flows to the load than is returned from it. If P is zero, all power sent to the load is returned.
- Thus, if P has a positive value, it represents the power that is really dissipated by the load. For this reason, P is called real power.
- In modern terminology, real power is also called active power. Thus, active power is the average value of the instantaneous power, and the terms real power, active power, and average power mean the same thing.

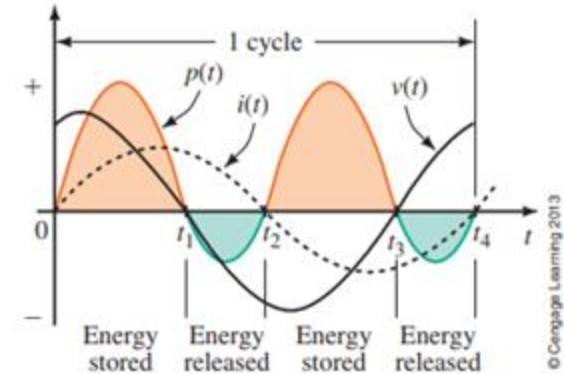


FIGURE 17-2 Instantaneous power in an ac circuit. Positive p represents power to the load; negative p represents power returned from the load.

Reactive Power

- Although oscillating power does no useful work, it cannot be ignored.
- Extra current is required to create this oscillating power, and this current must be supplied by the source; this also means that conductors, circuit breakers, switches, transformers, and other equipment must be made physically larger to handle the extra current. This increases the cost of a system.
- We define this oscillating power as Reactive power.

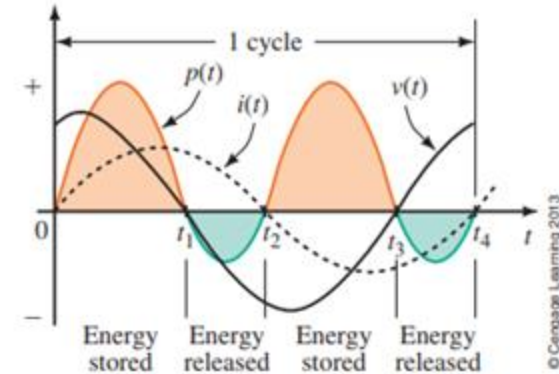


FIGURE 17-2 Instantaneous power in an ac circuit. Positive p represents power to the load; negative p represents power returned from the load.

Active and Reactive Power

- During the intervals that p is negative, power is being returned from the load. This can only happen if the load contains reactive elements: L or C.
- The portion of power that flows into the load then back out is called reactive power.
- Since it first flows one way then the other, its average value is zero; thus, reactive power contributes nothing to the average power to the load.
- The symbol for reactive power is Q , and its unit of measure is the volt-ampere reactive (VAR).

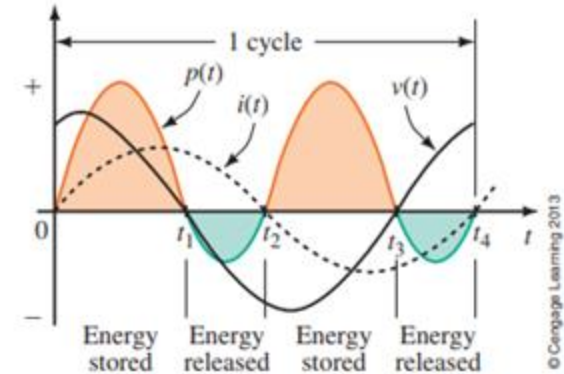


FIGURE 17-2 Instantaneous power in an ac circuit. Positive p represents power to the load; negative p represents power returned from the load.

Active and Reactive Power

Instantaneous
Power

$$p = VI \cos \theta (1 - \cos 2\omega t) + VI \sin \theta \sin(2\omega t)$$

Average
Power

$$P = VI \cos \theta$$

Reactive
Power

$$Q = VI \sin \theta$$

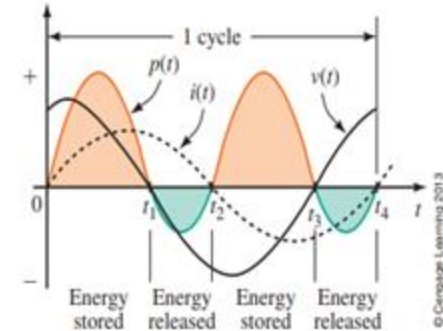


FIGURE 17-2 Instantaneous power in an ac circuit. Positive p represents power to the load; negative p represents power returned from the load.

Note that real power and reactive power are not separate “types” of power, they are merely different components of the total power

Apparent Power and Power Factor

- When a load has voltage V across it and current I through it as in Figure 17–11, the power that appears to flow to it is VI .
- However, if the load contains both resistance and reactance, this product represents neither real power nor reactive power.
- Since VI appears to represent power, it is called **apparent power**. Apparent power is given the symbol S and has units of volt-amperes (VA).

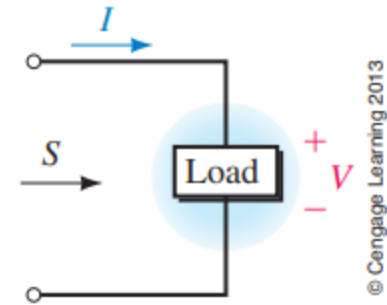


FIGURE 17–11 Apparent power $S = VI$.

Apparent Power and Power Factor

Equipment Is Rated in
VA



FIGURE 17-12 Power apparatus is rated in apparent power. The transformer shown is a 167-kVA unit.

Apparent Power and Power Factor

- The quantity $\cos \theta$ is defined as power factor and is given the symbol F_p or p.f..
- Power factor is expressed as a number or as a percent. is apparent that the power factor cannot exceed 1.0 (or 100%).

Instantaneous
Power

$$p = VI \cos \theta (1 - \cos 2\omega t) + VI \sin \theta \sin(2\omega t)$$

Power
Factor

$$F_p = \cos \theta$$

Note: V and I are the rms values

Apparent Power and Power Factor

- A load's power factor shows how much of its apparent power is actually real power.

Power
Factor

$$F_p = \cos \theta$$

Average
Power

$$P = VI \cos \theta$$

Load Type	θ (degree)	Power Factor
Resistive Load	0	1, referred to as unity power factor
Pure Inductive Load	90	0
Pure Capacitive Load	-90	0
Primarily Inductive	+ve (somewhere between 0 and 90)	$0 < f_p < 1$, described as lagging
Primarily Capacitive	-ve (somewhere between 0 and -90)	$0 < f_p < 1$ described as leading

Note: V and I are the rms values

Apparent Power and Power Factor

- Let's assume $V=220\text{ V}$ and $I=1.5\text{ A}$.

Power Factor $F_p = \cos \theta$

Average Power $P = VI \cos \theta$

θ (degree)	Real Power (W)
0	330
25	299.08
50	212.12
75	85.41
90	0

Note: V and I are the rms values

Power Triangle

- The three quantities average power, apparent power, and reactive power can be related in the **vector domain**.
- Thus, it is a standard practice to represent S, P, and Q in the form of a triangle, known as the **power triangle**.
- If a network has both capacitive and inductive elements, the reactive component of the power triangle will be determined by the **difference between the reactive power delivered to each**. If Q_L is greater than Q_C , the resultant power triangle will be similar to Fig. 14. If Q_C is greater than Q_L , the resultant power triangle will be similar to Fig. 15.

Apparent
Power

$$S = P + jQ$$

Net Reactive
Power

$$Q = Q_L - Q_C$$

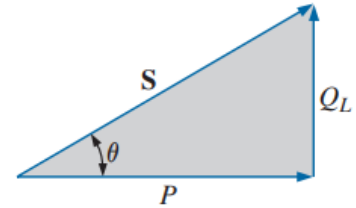


FIG. 14

Power diagram for inductive loads.

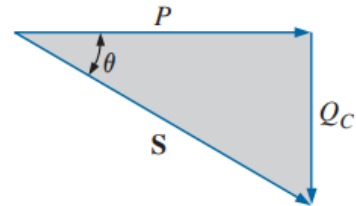


FIG. 15

Power diagram for capacitive loads.

Power Triangle

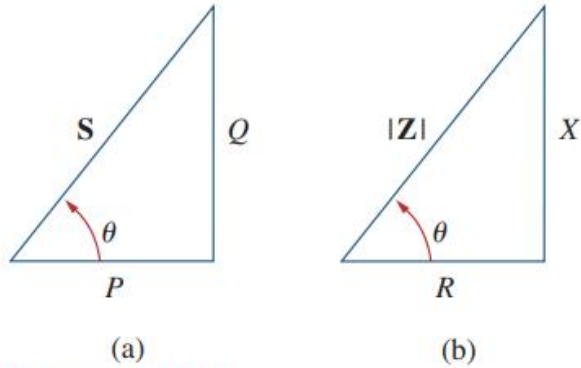


Figure 11.21

(a) Power triangle, (b) impedance triangle.

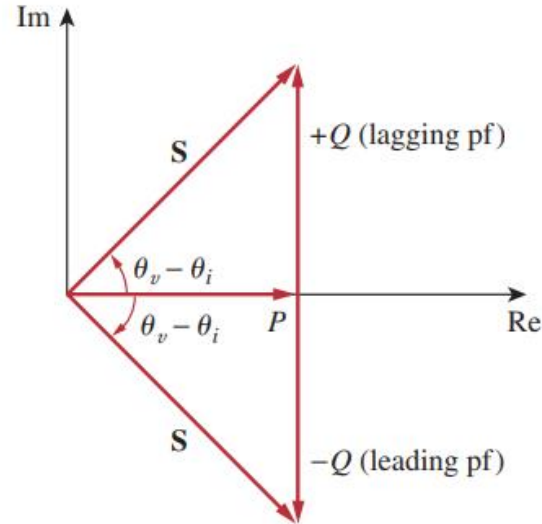


Figure 11.22

Power triangle.

$$P = I^2 R = \frac{V^2}{R}$$

$$Q_L = I^2 X_L = \frac{V^2}{X_L}$$

$$Q_C = I^2 X_C = \frac{V^2}{X_C}$$

Power Triangle

$$\text{Complex Power} = \mathbf{S} = P + jQ = \mathbf{V}_{\text{rms}}(\mathbf{I}_{\text{rms}})^*$$

$$= |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| \angle \theta_v - \theta_i$$

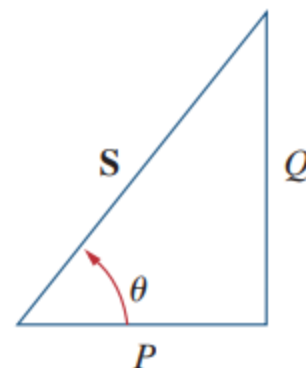
$$\text{Apparent Power} = S = |\mathbf{S}| = |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| = \sqrt{P^2 + Q^2}$$

$$\text{Real Power} = P = \text{Re}(\mathbf{S}) = S \cos(\theta_v - \theta_i)$$

$$\text{Reactive Power} = Q = \text{Im}(\mathbf{S}) = S \sin(\theta_v - \theta_i)$$

$$\text{Power Factor} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

Power Triangle



$$\theta = \theta_v - \theta_i$$

$$Q = P \tan \theta$$

Example #1

Find the total number of **watts**, **volt-amperes reactive**, **volt-amperes**, and **power factor** of the network. Draw the **power triangle** and find the **current in phasor form**.

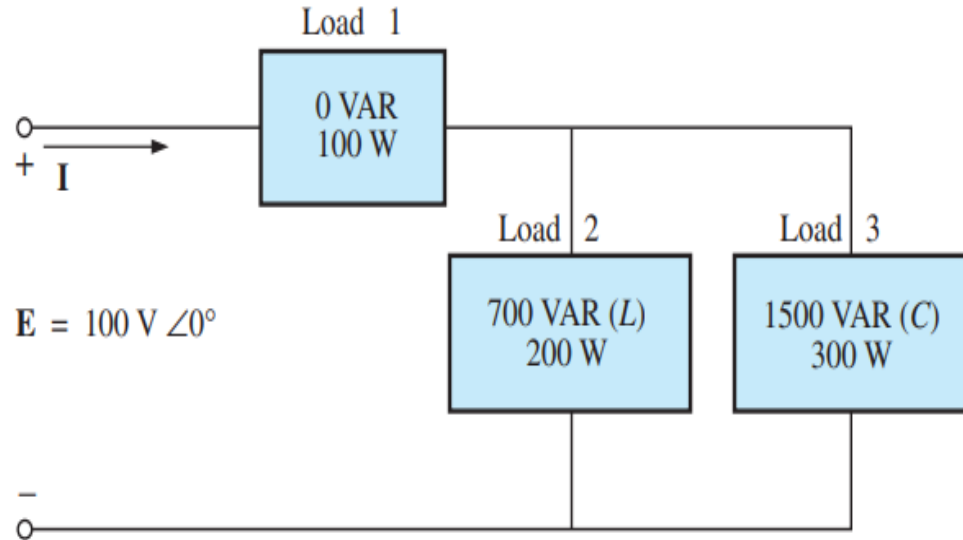


Figure for Example #1

Examples #2

A load Z draws **12 kVA** at a power factor of **0.856 lagging** from a **120-V rms** sinusoidal source. Calculate: (a) the average and reactive powers delivered to the load, (b) the peak current, and (c) the load impedance.

Power Factor Correction

- The design of any power transmission system is very sensitive to the magnitude of the current in the lines as determined by the applied loads. **Increased currents result in increased power losses** (by a squared factor since $P = I^2 * R$) in the transmission lines due to the resistance of the lines.
- Heavier currents also require larger conductors, increasing the amount of copper needed for the system, and, quite obviously, they require increased generating capacities by the utility company.
- **Every effort must therefore be made to keep current levels at a minimum. Since the line voltage of a transmission system is fixed, the apparent power is directly related to the current level.** In turn, the smaller is the net apparent power, the smaller is the current drawn from the supply. Minimum current is therefore drawn from a supply when $S = P$ and $QT = 0$.

Power Factor Correction

- The process of increasing the power factor without altering the voltage or current to the original load is known as **power factor correction**. Alternatively, the process of introducing reactive elements to bring the power factor closer to unity is called **power-factor correction**.

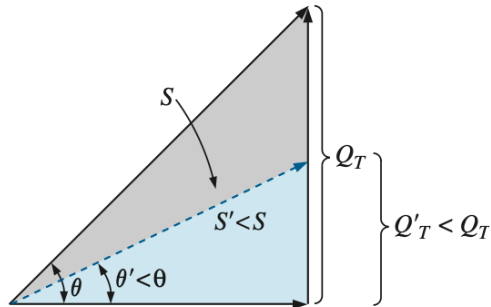


FIG. 28

Demonstrating the impact of power-factor correction on the power triangle of a network.

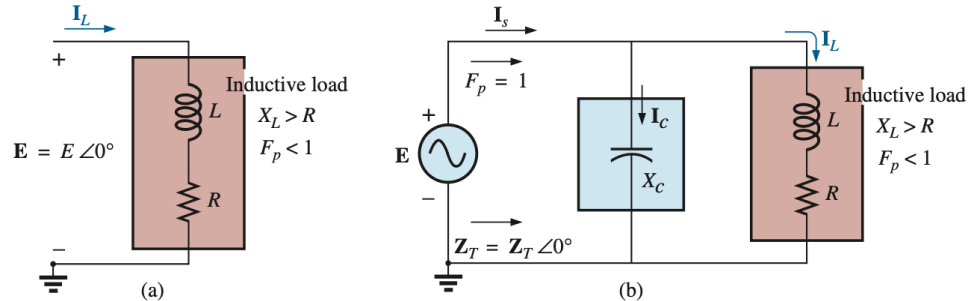


FIG. 29

Demonstrating the impact of a capacitive element on the power factor of a network.

Examples #3

A **5 hp** motor with a **0.6 lagging power factor** and an **efficiency of 92%** is connected to a **208 V, 60 Hz supply**.

- a) Establish the **power triangle** for the load.
- b) Determine the **power-factor capacitor** that must be placed in parallel with the load to raise the power factor to unity.
- c) Determine the **change in supply current** from the uncompensated to the compensated system.

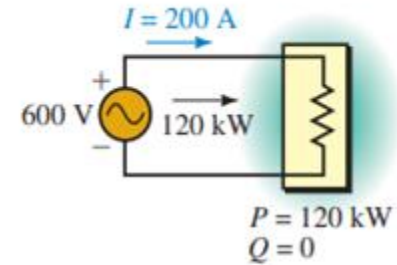
Examples #4

A small industrial plant has a **10-kW heating load** and a **20 kVA inductive load** due to a bank of induction motors. The heating elements are considered **purely resistive ($Fp = 1$)**, and the induction motors have a **lagging power factor of 0.7**. The supply is **1000 V at 60 Hz**.

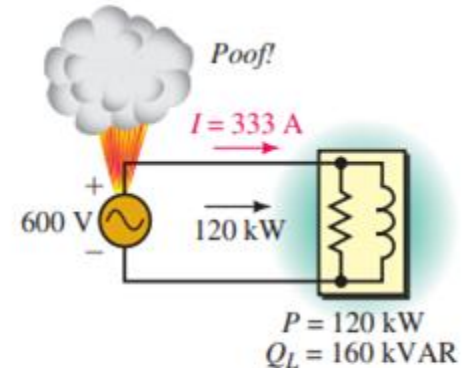
- a. Establish the **power triangle** for the load.
- b. Determine the capacitive element required to **raise the power factor to 0.95**.
- b. Compare the **levels of current drawn** from the supply.

Why Equipment Is Rated in VA

- Assume that the generator is rated at 600 V, 120 kVA. This means that it is capable of supplying $I = 120 \text{ kVA} / 600 \text{ V} = 200 \text{ A}$.
- In (a), the generator is supplying a purely resistive load with 120 kW. Since $S = P$ for a purely resistive load, $S = 120 \text{ kVA}$ and the generator is supplying its rated current.
- In (b), the generator is supplying a load with $P = 120 \text{ kW}$ as before, but $Q = 160 \text{ kVAR}$. Its apparent power is therefore $S = 200 \text{ kVA}$, which means that the generator current is $I = 200 \text{ kVA} / 600 \text{ V} = 333.3 \text{ A}$.
- Even though it is supplying the same power as in (a), the generator is now greatly overloaded, and damage may result as indicated in (b).
- This example illustrates clearly that rating a load or device in terms of power is a poor choice, as its current-carrying capability can be greatly exceeded (even though its power rating is not).
- Thus, the size of electrical apparatus (generators, interconnecting wires, transformers, etc.) required to supply a load is governed, not by the load's power requirements, but rather by its VA requirements



(a) $S = 120 \text{ kVA}$



$$(b) S = \sqrt{(120)^2 + (160)^2} = 200 \text{ kVA}$$

The generator is overloaded